

ADAS-TO × TabPFN 3.5

Forecasting post-takeover behaviour from pre-takeover signals

Driver-disjoint evaluation · TabPFN + tree baselines · feature ablations

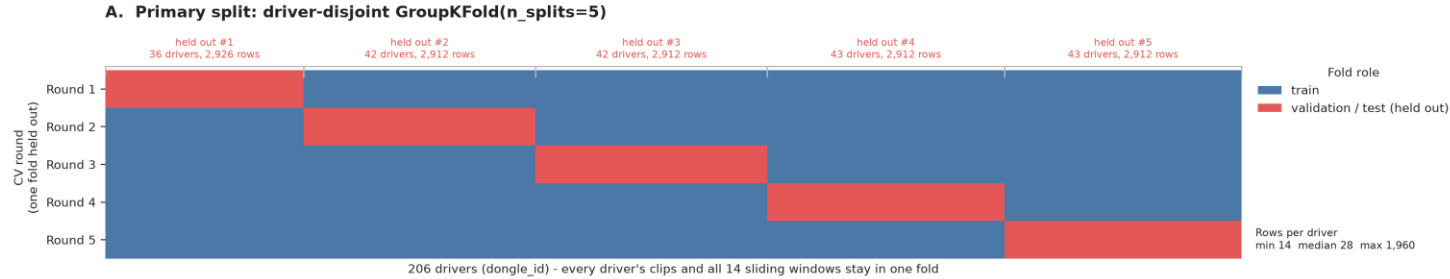
Investigation by Gabriel Amine-Eddine

What we did

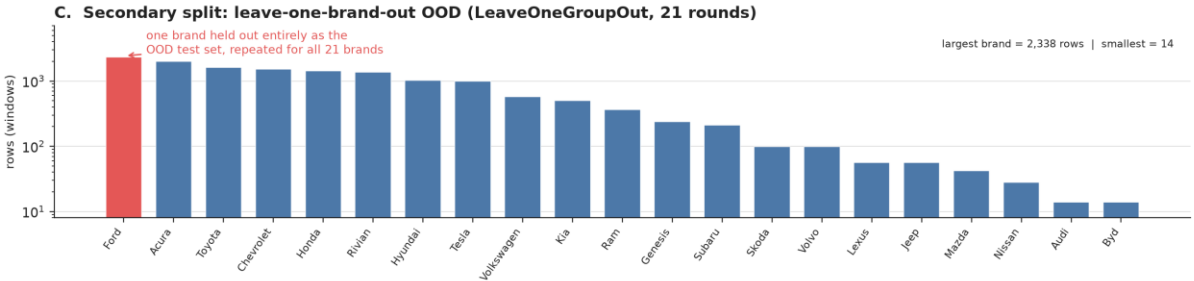
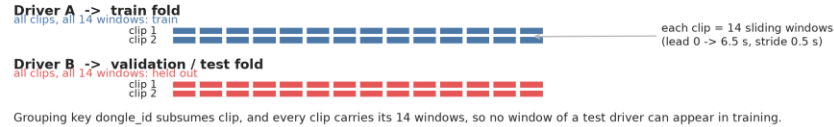
- Task — from the pre-takeover window $[-5, 0]$ s, predict the post-takeover maneuver (`post_maneuver_type`) and its intensity (steer torque, jerk).
- Data — ADAS-TO sample dataset (<https://huggingface.co/datasets/HenryYHW/ADAS-TO>): 1,591 clips, 1,043 re-identified/labelled, 208 drivers, 21 brands.
- Split — driver-disjoint train/val/test (the paper's protocol); brand-OOD reported separately.
- Models — TabPFN (in-context) vs LightGBM / XGBoost / CatBoost.
- Three feature families tested: CAN signals, YOLOv8n vision geometry, optical-flow residual.
- Deliverable — this deck + an explorer app that scores held-out test clips live.

The split: driver-disjoint, balanced by clip count

Train / validation split of the ADAS-TO sliding forecast dataset
14,574 rows · 206 drivers · 1,041 clips · 21 brands · no fixed test file: every row is scored once as held-out

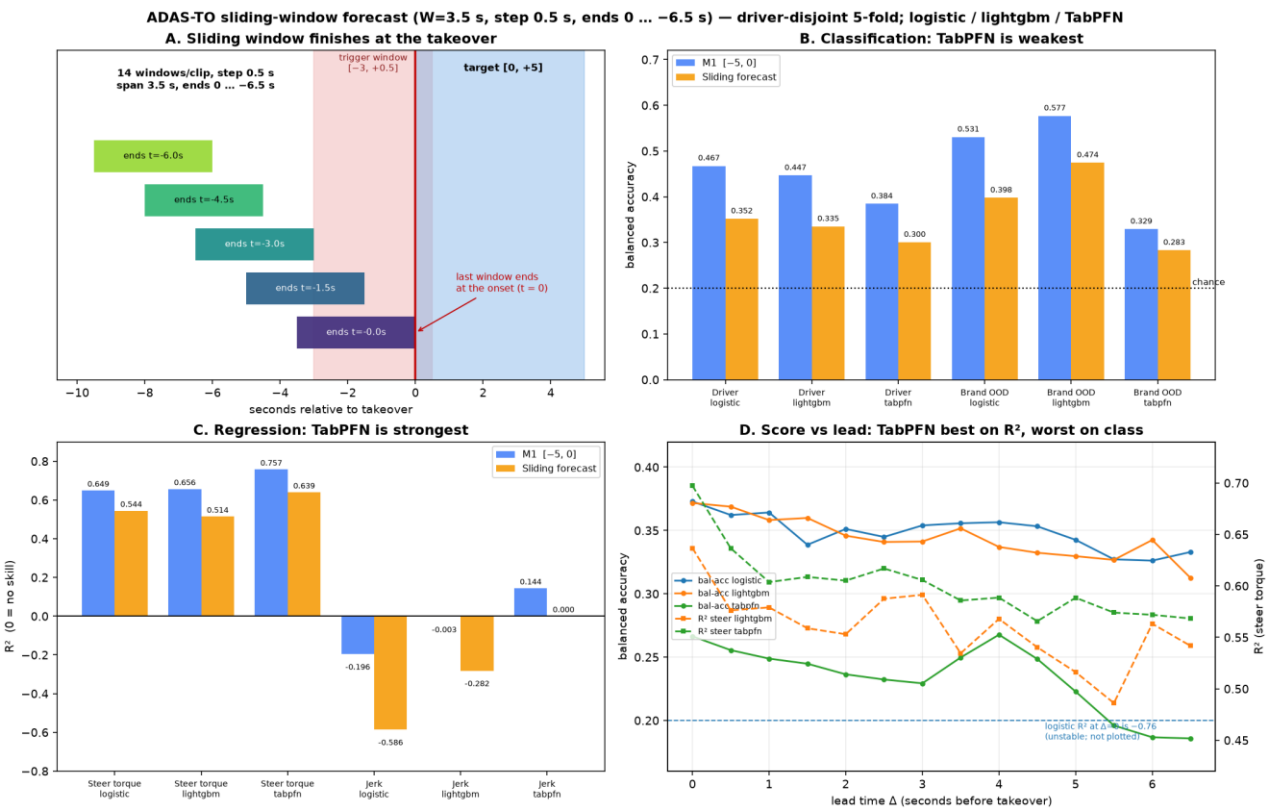


B. Why it is grouped by driver: one driver's whole history lands in a single split



- Group key = driver (dongle_id). All of a driver's clips stay in one split.
- Why: 208 drivers, median 2 clips, top driver = 13.4% of rows — clip balancing
- stops one driver skewing a split.
- Fixed split: train 625 / val 209 / test 209 rows.
- Honest scoring: test clips are never seen by the model that scores them.

Experiment 1 — CAN-only baseline (frozen for ablations)

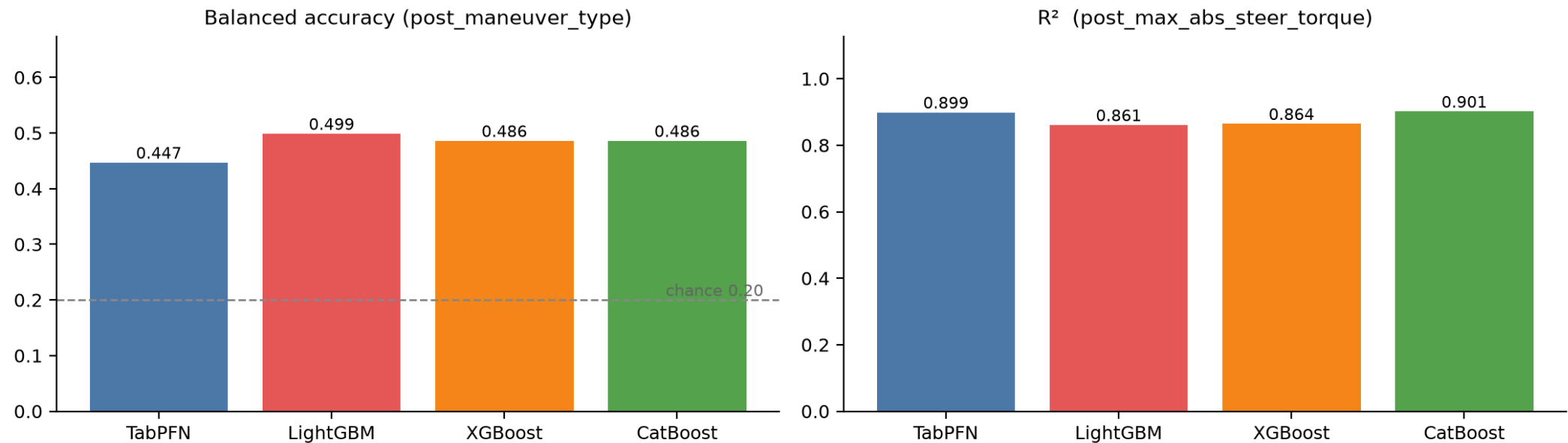


metric	lightgbm	tabpfn
maneuver bal-acc	0.335	0.300
macro-F1	0.306	0.244
steer-torque R^2	0.514	0.639
jerk R^2	-0.282	0.000

- Driver-disjoint CV on the sliding forecast table (14,574 windows).
- Steer-torque $R^2 \approx 0.51$ – 0.64 is the real signal: pre-window kinematics predict post-takeover steering intensity.
- Maneuver class is weak (bal-acc 0.30–0.34 vs 0.25 chance); jerk is not predictable ($R^2 \approx 0$).
- This is the CAN baseline both feature ablations are diffed against.

Experiment 2 — four models on the test split

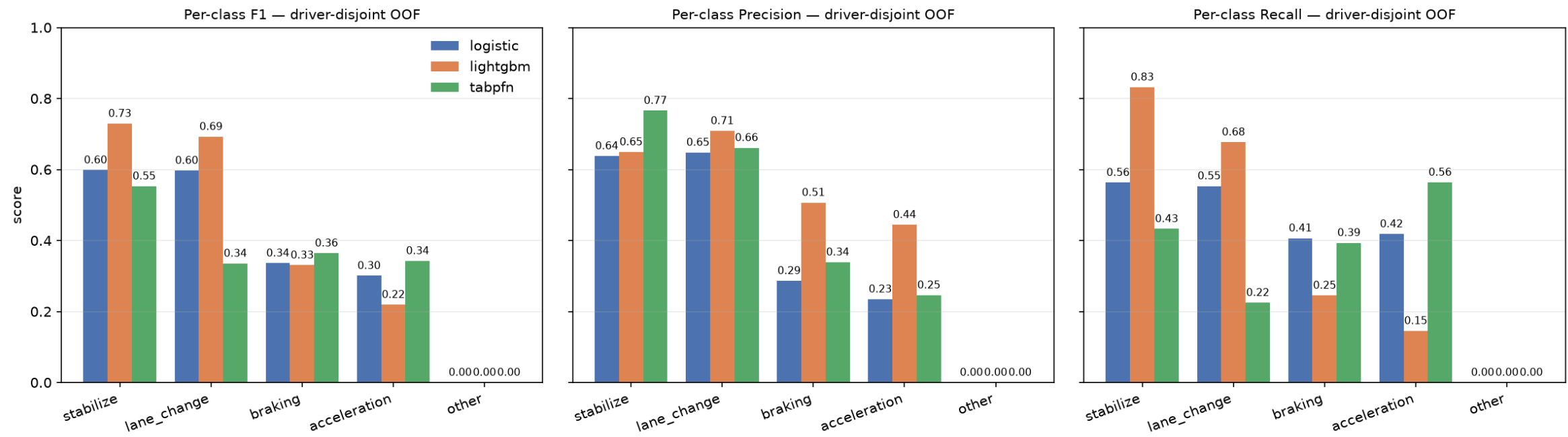
Four models, driver-disjoint test split (209 clips)



model	bal-acc	macro-F1	accuracy	steer R²
tabpfn	0.447	0.378	0.388	0.899
lightgbm	0.499	0.498	0.694	0.861
xgboost	0.486	0.484	0.684	0.864
catboost	0.486	0.485	0.684	0.901

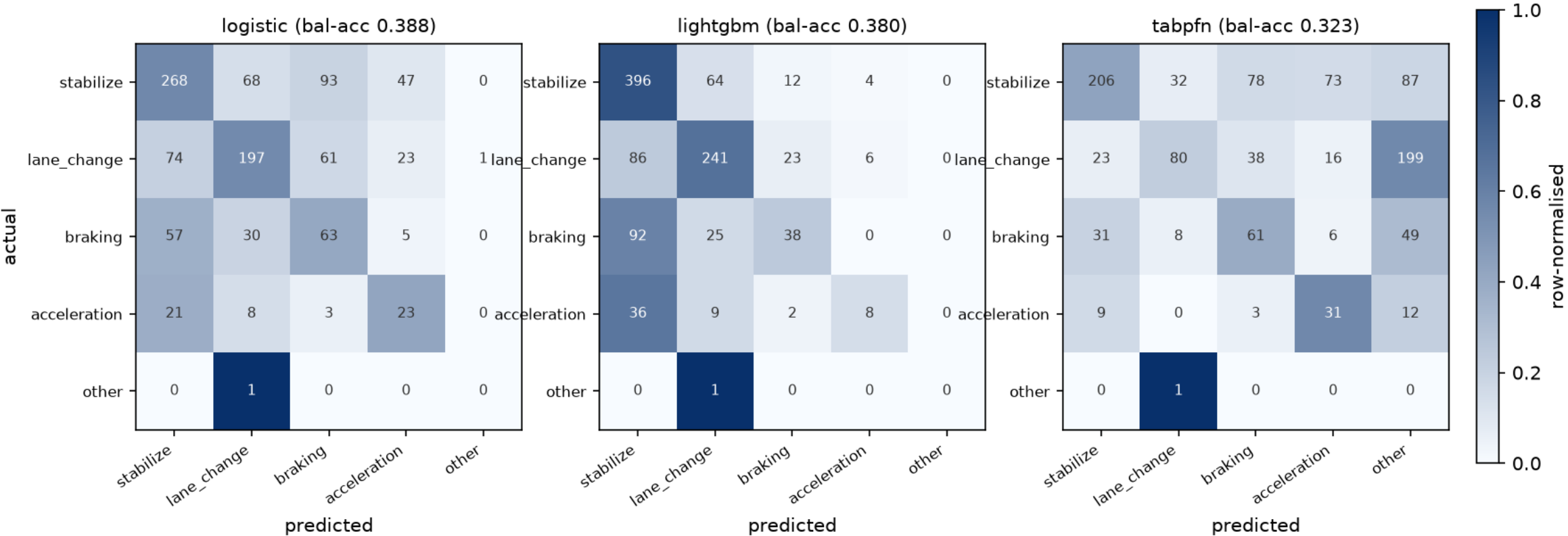
Per-class behaviour

post_maneuver_type — class-by-class scores (pooled out-of-fold, 5-fold GroupKFold by driver)

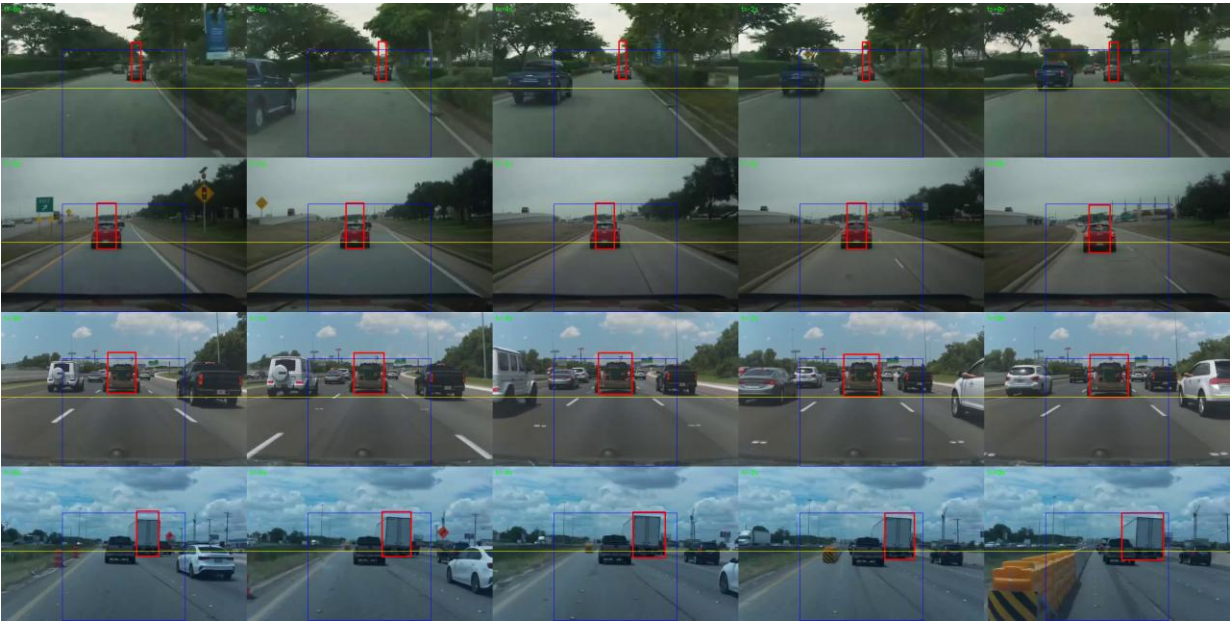


Per-class behaviour contd.

Confusion matrices — post_maneuver_type, driver-disjoint OOF



Experiment 3 — YOLOv8n vision features (null result)



metric	CAN	CAN+vision	CAN+light
maneuver bal-acc (tabpfn)	0.300	0.305	0.306
steer R ² (lightgbm)	0.514	0.533	0.546
bal-acc brand-OOD (lgbm)	0.474	0.433	0.467

- alignment QA: median Spearman 0.75, 83% positive — PASS
- The detector is not the excuse: it matches radar's lead in 94% of lead windows.
- Net gain is noise, and it reverses out-of-distribution (brand-OOD).
- Read as a clean null, not a failure of the model.

Experiment 4 — optical-flow features (null result)



metric	CAN	flow only	CAN+flow
maneuver bal-acc (lgbm)	0.335	0.290	0.334
steer R ² (lgbm)	0.514	-0.409	0.547
bal-acc brand-OOD (lgbm)	0.474	0.350	0.460

- alignment QA: median Spearman -0.026, 55% negative — FAIL (radar cross-check)
- Pooled bump does not survive the non-radar / brand-OOD test.
- Residual is dominated by ego motion (largest blob $\approx$ 12% of frame).
- Verdict: third feature family, third null result.

The explorer — four models, live, on held-out clips



- Shows test-split clips only.
- Each column = one model, filled asynchronously as its prediction lands.
- Per-model timing shown; TabPFN uses `fit_with_cache`.
- Top-left score cards = batch test-set metrics.

Takeaways

- Pre-takeover CAN signals genuinely predict post-takeover steering intensity (steer $R^2 \approx 0.51$ – 0.66); the maneuver class is only weakly predictable.
- Expectation of better performance metrics with full ADAS-TO Dataset
- On the held-out test split: LightGBM leads the class task (bal-acc 0.499); CatBoost / TabPFN lead intensity regression (R^2 0.90).
- Vision geometry and optical flow both add nothing net once the split is cross-platform — pooled gains reverse under brand-OOD.
- Guardrails matter: driver-disjoint splits, no trigger leakage, honest metrics.
- Reproduce with `scripts/run_experiments.py` and `scripts/score_test.py`; explore with `app/server.py`.